PRD: Passkeys for card checkout in India

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Offer a passkey right after a successful OTP card payment, so the shopper's next card payment takes one fingerprint instead of an SMS wait. A concept PRD for my portfolio that builds on my Mastercard checkout teardown; not affiliated with Mastercard.

Problem

Nearly 35% of payment failures in India start at authentication: OTPs that arrive late, OTPs typed wrong and redirects that break (Razorpay, March 2026). UPI has no such step, and shoppers feel the difference.

Way to pay ₹2,499 online	Steps	Characters typed	Leaves the checkout	Waits for an SMS
New card, with OTP	7	29	Yes, to the bank page	Yes
Saved card, with OTP	4	6–9	Yes, to the bank page	Yes
UPI	3	4–6	Yes, to the UPI app	No
Saved card, with passkey (this PRD)	2	0	No	No

Counts come from my teardown of a typical Android checkout with a tokenised domestic card; they vary by merchant, gateway and bank.

Cards still carry the money that matters most online: ₹1.23 trillion of the ₹2.02 trillion spent on credit cards in August 2026 was online (Angel One). Each failed authentication puts a big, often credit-funded order at risk.

Why now

In 2026 the rules, the rails and the price gap all moved: RBI accepts biometrics, passkey checkout is live in India, and big UPI payments start costing merchants on 15 October.

Date	What happened	Why it matters here
15 Oct 2026	UPI payments to merchants above ₹2,000 start carrying a 0.4% fee, capped at ₹300; smaller payments, over 95% of UPI merchant volume, stay free (Deccan Chronicle)	Big UPI payments stop being free for merchants, in the big-ticket space where cards compete
1 Oct 2026	Card issuers must run risk-based checks on cross-border card-not-present payments (Entrackr)	Issuers are already rebuilding authentication this year
1 Apr 2026	RBI's authentication directions take effect: two factors per payment, one of them dynamic; biometrics and device-based tokens join SMS OTP; issuers fully compensate losses from non-compliance (Entrackr)	A fingerprint on a bound device now counts, with no new regulation needed
31 Mar 2026	Razorpay launches biometric passkey card authentication with Mastercard and Visa, reporting up to 95% transaction success (Razorpay)	The rails work in India, in live checkouts
29 Aug 2024	Mastercard picks India for the global launch of its Payment Passkey Service, piloting with Juspay, Razorpay, PayU, bigbasket and Axis Bank (Mastercard)	The standard and the first partners already exist

Goals and non-goals

The goal is fewer lost card payments from returning shoppers, with fraud no higher than today. Targets are for a 90-day pilot on one gateway; they are my estimates, to be reset once real baselines are in.

Goal	Measure	Pilot target
Move returning card shoppers off SMS OTP	Enrolments ÷ enrolment offers shown	30%
Make passkey payments succeed first time	Successful ÷ attempted passkey payments	95%, Razorpay's reported benchmark

Goal	Measure	Pilot target
Make paying faster	Median time from tapping Pay to confirmation	Under 10 seconds
Keep fraud flat	Fraud and chargeback rates, passkey vs OTP payments	No higher

Not in scope:

- Removing OTP: it stays for first payments, fallbacks and high-risk step-ups.
- Replacing banks' authentication pages; a standard in-app screen is a separate bet.
- Cross-border card payments, UPI and debit card PIN flows in phase 1.
- Syncing passkeys across a shopper's devices; phase 1 passkeys are bound to one phone.

Users

The users are returning shoppers who pay big amounts online by credit card. UPI does not own these payments: over 95% of UPI merchant payments are ₹2,000 or less (Deccan Chronicle).

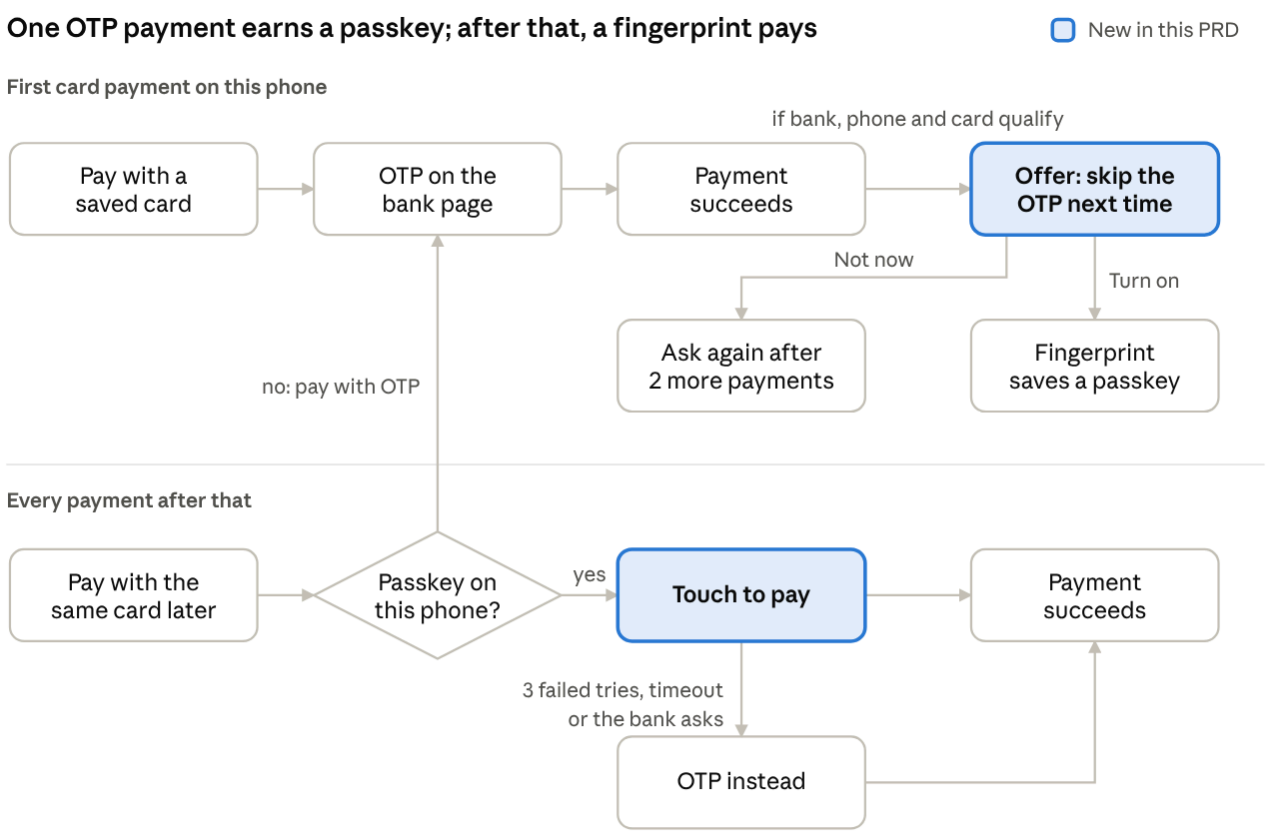
Shopper	Why they pay by card	What they need from checkout
Big-ticket buyer	No-cost EMI on a phone, laptop or flight	The payment to go through first time, at a high amount
Rewards maximiser	Points, cashback and lounge access on every spend	Speed on repeat buys from saved cards
New-to-credit cardholder	A first card, still learning what is safe	Reassurance, with OTP one tap away

The base keeps growing: 124.05 million credit cards in August 2026, up from 112.34 million a year earlier (Angel One).

Job story: when I pay a big amount by card on my phone, I want to approve it in one touch, so I don't lose the order waiting for an OTP.

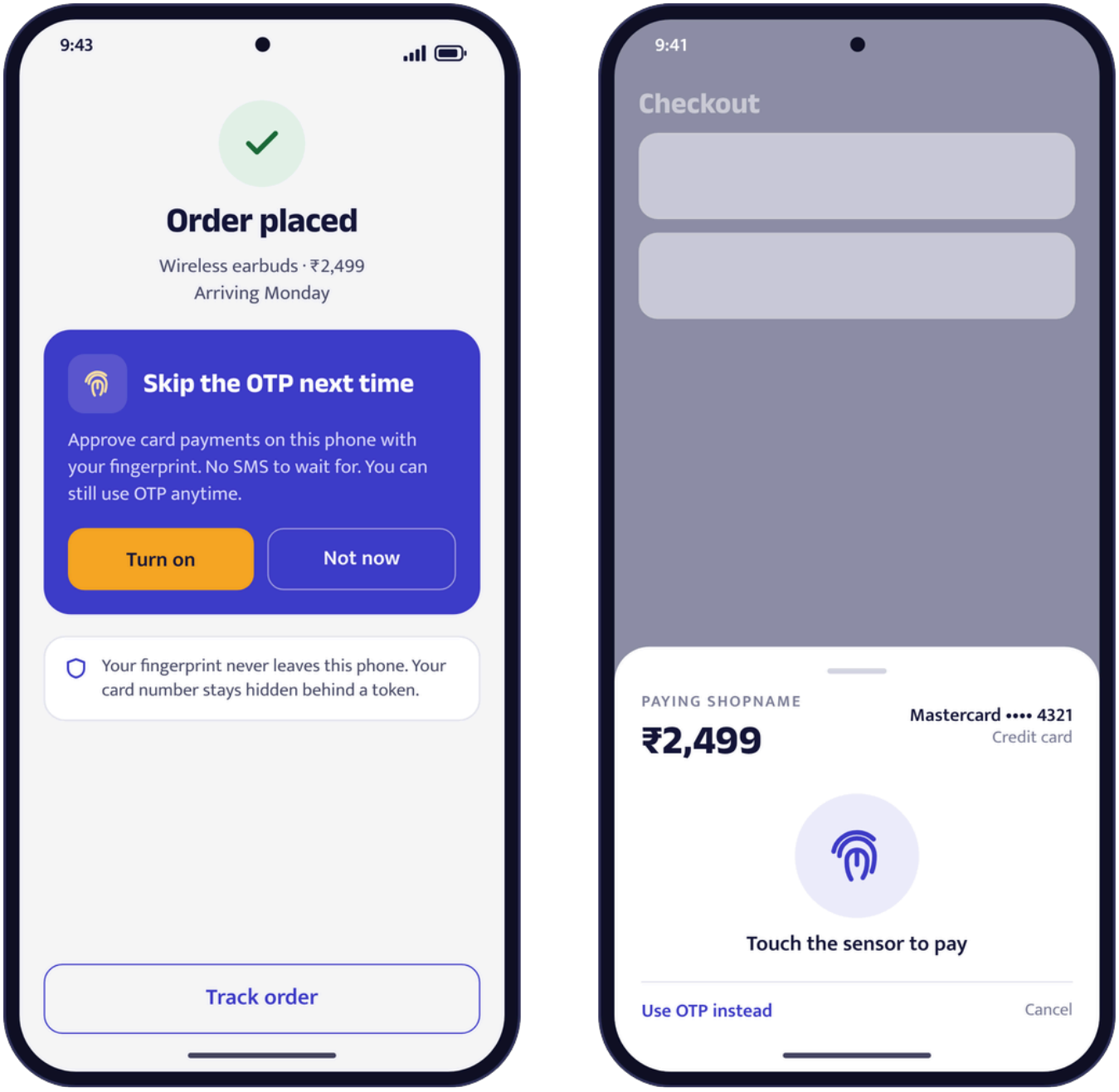
Solution

Ask when trust is highest, right after an OTP payment succeeds. From then on, the saved card is approved with a fingerprint, and OTP stays one tap away.



passkey flow · first payment, then every payment after

The offer appears only after a payment has worked. A missing passkey, a failed fingerprint or a bank's risk check always falls back to the OTP shoppers already know.



My concept screens from the teardown: the offer on the order-placed screen (left) and the next checkout (right). They are my designs, not Mastercard's.

Requirements

Twelve requirements, eight of them P0 for the pilot. The gateway SDK carries most of the work, and OTP stays the safety net throughout.

ID	Requirement	Built by	Priority
R1	Offer passkey set-up on the success screen of an OTP-verified card payment, never before paying	Gateway SDK, merchant	P0

ID	Requirement	Built by	Priority
R2	Create the passkey only on the phone that just completed the OTP payment, bound to the card's token	Mastercard, issuer	P0
R3	Show "Pay with fingerprint" on saved cards that have a passkey on this phone	Gateway checkout	P0
R4	Keep "Use OTP instead" visible on every passkey prompt	Gateway checkout	P0
R5	Fall back to OTP after 3 failed biometric tries, a 30-second timeout or an issuer step-up	Gateway, issuer	P0
R6	Offer set-up only when the issuer supports passkeys, the phone has a screen lock and biometric, and the card is tokenised	Gateway SDK	P0
R7	Tell the cardholder by SMS and in the bank app when a passkey is added or removed	Issuer	P0
R8	Log offer shown, set-up started and finished, passkey success or failure, and every fallback	Gateway	P0
R9	Say in one line that the fingerprint stays on the phone and the card number stays hidden	Merchant, gateway	P1
R10	Ask at most 3 times per card per phone; "Not now" waits for 2 more OTP payments	Gateway SDK	P1
R11	Let cardholders see and remove passkeys, phone by phone, in the bank app	Issuer	P1
R12	Set-up and checkout copy in Hindi and four regional languages	Merchant, gateway	P2

P0 is needed for the pilot, P1 before scaling to all merchants, P2 after.

Edge cases

Every edge case ends in a working payment: when a passkey can't be used, the shopper gets today's OTP flow, never a dead end.

Case	What happens
New phone or app reinstalled	No passkey on this phone, so the payment uses OTP; set-up is offered again once it succeeds
Shared family phone	Set-up warns that anyone whose fingerprint opens this phone can approve payments, and offers “Not on this phone”
Fingerprint fails 3 times	Switches to OTP for this payment; the passkey stays
No screen lock or no biometric	Set-up is never offered
Card replaced or token deleted	The passkey stops working; the next payment uses OTP and offers set-up again
Phone lost or stolen	The cardholder removes passkeys in the bank app; blocking the card removes them too
Issuer flags the payment as risky	The issuer asks for an OTP on top, even with a passkey
Network drops during approval	Checkout retries once, shows “Checking your payment” and never charges twice
Merchant or bank without passkey support	The standard OTP flow, with nothing broken or greyed out

Security and compliance

A passkey payment meets RBI’s two-factor rule by itself: the phone holding the passkey is one factor, the fingerprint or face is the other, and each approval is unique to that payment.

- **RBI’s 2026 directions:** at least two factors per payment, one of them dynamic; biometrics and device-based tokens are allowed alongside SMS OTP (Entrackr). Issuers may let customers choose their factor (Khaitan & Co), which “Use OTP instead” does.
- **Biometrics stay on the phone:** matching happens on the device and is never sent to a server; the bank gets only a signed approval. Passkeys are phishing resistant, so a fake payment page can’t reuse one (FIDO Alliance).
- **Card number stays hidden:** the passkey works with the card’s token, so no account data is shared with third parties (Mastercard).

- **Safe set-up:** a passkey is created only on the phone that just passed an OTP payment, and the cardholder is told every time one is added (R2, R7).
- **Liability:** issuers must fully compensate customers for losses from non-compliance (Entrackr), so each pilot issuer signs off the risk rules before launch.
- **Data:** no new personal data is collected, and the flow stays within the Digital Personal Data Protection Act, 2023.
- **Access:** a device PIN or OTP remains for anyone who can't use biometrics.

Metrics

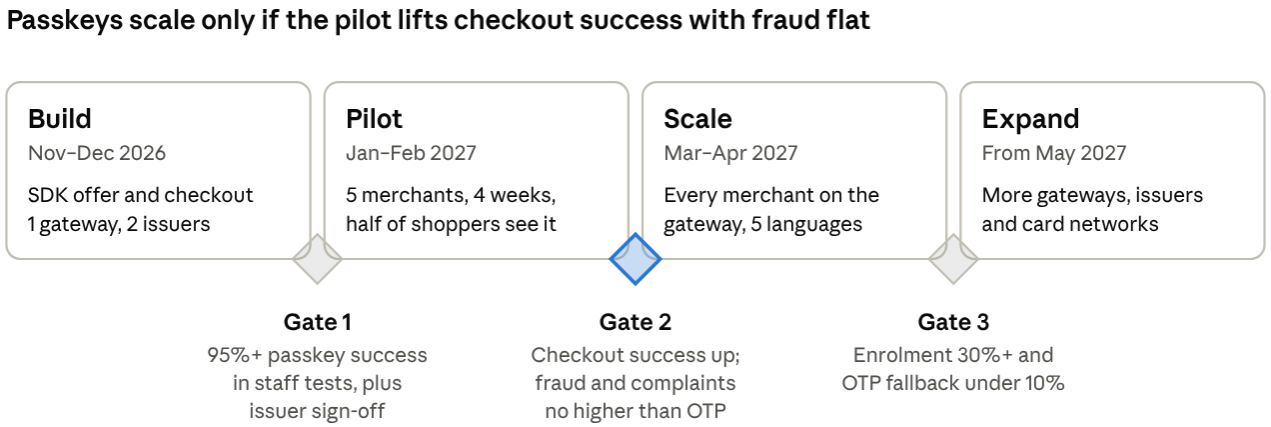
The north star is card checkout success: completed card payments ÷ card payments started. Every other metric explains a move in it or guards against a bad one.

Metric	Definition	Pilot target
North star: card checkout success	Completed ÷ started card payments, enrolled shoppers	+5 points vs the OTP control group
Enrolment rate	Set-ups finished ÷ offers shown	30%
Passkey share	Passkey approvals ÷ all card payments by enrolled shoppers	70% by day 60
Passkey success	Successful ÷ attempted passkey approvals	95%
Authentication drop-off	Payments abandoned at the authentication step	35% lower than OTP, in line with the cut in authentication errors Razorpay reports
Time to pay	Median seconds from tapping Pay to confirmation	Under 10
Guardrail: fallback to OTP	Passkey attempts that end in OTP	Under 10%
Guardrail: fraud and chargebacks	Basis points of passkey payment value	No higher than OTP
Guardrail: "debited but failed" complaints	Complaints per 10,000 payments	No higher than OTP

Test: for 4 weeks on one gateway, half of eligible shoppers see the set-up offer and half don't. Scale up only if checkout success rises and every guardrail holds.

Rollout

Roll out in four phases, each opened by a gate, so passkeys reach every merchant only after the pilot proves them.



rollout · 4 phases, 3 gates

Each gate reuses the pilot targets in Metrics; a missed gate means fix and re-test, not scale. Dates assume work starts in November 2026.

Risks and mitigations

The biggest risk is adoption, not technology: shoppers and issuers both default to the OTP they know.

Risk	Likelihood	Mitigation
Few shoppers set one up, because OTP feels familiar and safe	High	Ask right after a payment works, with one line on why it's safe; cap at 3 asks (R1, R9, R10)
Issuers are slow to support passkeys	High	Start with the issuer from Mastercard's 2024 pilot, Axis Bank, and gateways that already run issuer authentication
Someone else's fingerprint on a shared phone approves a payment	Medium	A warning at set-up, issuer risk checks and removal from the bank app

Risk	Likelihood	Mitigation
A fraudster talks a victim into setting up a passkey on the fraudster's phone	Medium	Set-up only on the phone that just paid with OTP, and an alert for every new passkey (R2, R7)
Credit on UPI pulls big-ticket payments away from cards	Medium	Outside this PRD; pair it with showing card benefits, like EMI and rewards, on carts above ₹2,000
Issuers read RBI's terms differently, since phrases like "capable of being proven" aren't precisely defined (Khaitan & Co)	Low	Legal sign-off from each pilot issuer before launch

Open questions

- [ ] Which issuers can support payment passkeys on their authentication servers by January 2027?
- [ ] Can Mastercard and Visa passkeys share one checkout flow, or does each network need its own?
- [ ] Should one passkey cover every saved card from the same bank on a phone, or one card at a time?
- [ ] Do issuers want an amount above which they still ask for an OTP?
- [ ] How do we spot a shared phone without collecting new personal data?

Sources

- Razorpay: biometric passkey launch with Mastercard and Visa, 31 March 2026
- Mastercard: India chosen for the global launch of the Payment Passkey Service, 29 August 2024
- Entrackr: RBI issues new rules for authentication of digital payments, September 2025
- Khaitan & Co: RBI authentication mechanisms directions, October 2025
- FIDO Alliance: Passkeys
- Deccan Chronicle: NPCI announces 0.4% merchant charge on UPI payments above ₹2,000, 15 September 2026
- Angel One: credit card spending in August 2026, 25 September 2026
- My Mastercard checkout teardown, for the step and character counts, October 2026